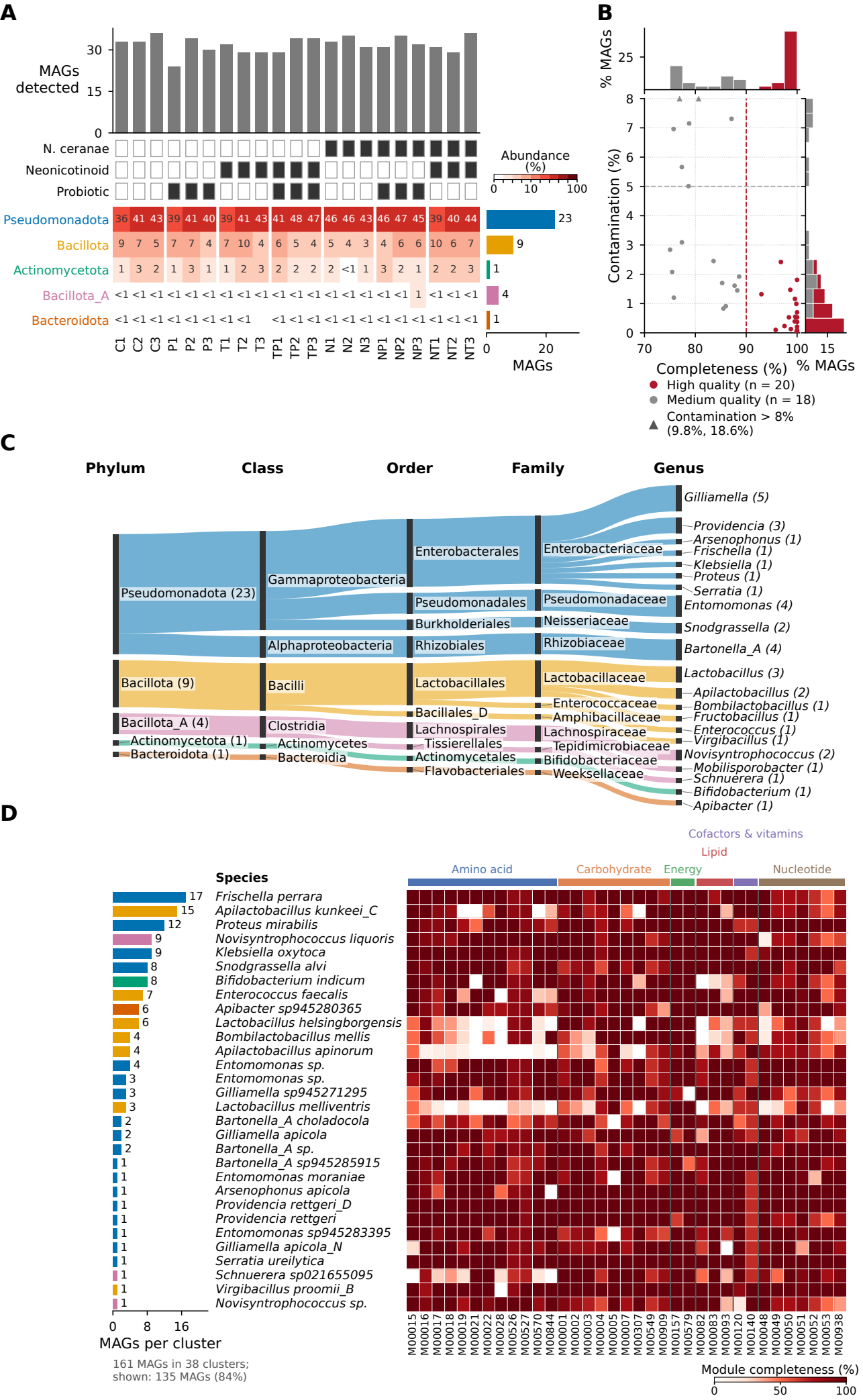

A
MAGs detected
30
15
0
N. ceranae
Neonicotinoid
Probiotic
Abundance (%)
0
10
100
Pseudomonadota 36 41 43 39 41 40 39 41 43 41 48 47 46 46 43 46 47 45 39 40 44
Bacillota 9 7 5 7 7 4 7 10 4 6 5 4 5 4 3 4 6 6 10 6 7
Actinomycetota 1 3 2 1 3 1 1 2 3 2 2 2 2 <1 1 3 2 1 2 2 3
Bacillota_A <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 1 <1 <1 <1
Bacteroidota <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1 <1
23
9
1
4
1
C1 C2 C3 P1 P2 P3 T1 T2 T3 TP1 TP2 TP3 N1 N2 N3 NP1 NP2 NP3 NT1 NT2 NT3
0
20
MAGs
B
% MAGs
25
Contamination (%)
8
7
6
5
4
3
2
1
0
70
80
90
100
15
% MAGs
Completeness (%)
High quality (n = 20)
Medium quality (n = 18)
Contamination > 8%
(9.8%, 18.6%)
C
Phylum
Class
Order
Family
Genus
Pseudomonadota (23)
Bacillota (9)
Bacillota_A (4)
Actinomycetota (1)
Bacteroidota (1)
Gammaproteobacteria
Alphaproteobacteria
Bacilli
Clostridia
Actinomycetes
Bacteroidia
Enterobacterales
Pseudomonadales
Burkholderiales
Rhizobiales
Lactobacillales
Bacillales_D
Lachnospirales
Tissierellales
Actinomycetales
Flavobacteriales
Enterobacteriaceae
Pseudomonadaceae
Neisseriaceae
Rhizobiaceae
Lactobacillaceae
Enterococcaceae
Amphibacillaceae
Lachnospiraceae
Tepidimicrobiaceae
Bifidobacteriaceae
Weeksellaceae
Gilliamella (5)
Providencia (3)
Arsenophonus (1)
Frischella (1)
Klebsiella (1)
Proteus (1)
Serratia (1)
Entomomonas (4)
Snodgrassella (2)
Bartonella_A (4)
Lactobacillus (3)
Apilactobacillus (2)
Bombilactobacillus (1)
Fructobacillus (1)
Enterococcus (1)
Virgibacillus (1)
Novisyntrophococcus (2)
Mobilisporobacter (1)
Schnuerera (1)
Bifidobacterium (1)
Apibacter (1)
D
Species
17 Frischella perrara
15 Apilactobacillus kunkeei_C
12 Proteus mirabilis
9 Novisyntrophococcus liquoris
9 Klebsiella oxytoca
8 Snodgrassella alvi
8 Bifidobacterium indicum
7 Enterococcus faecalis
6 Apibacter sp945280365
6 Lactobacillus helsingborgensis
4 Bombilactobacillus mellis
4 Apilactobacillus apinorum
4 Entomomonas sp.
3 Entomomonas sp.
3 Gilliamella sp945271295
3 Lactobacillus melliventris
2 Bartonella_A choladocola
2 Gilliamella apicola
2 Bartonella_A sp.
1 Bartonella_A sp945285915
1 Entomomonas moraniae
1 Arsenophonus apicola
1 Providencia rettgeri_D
1 Providencia rettgeri
1 Entomomonas sp945283395
1 Gilliamella apicola_N
1 Serratia ureilytica
1 Schnuerera sp021655095
1 Virgibacillus proomii_B
1 Novisyntrophococcus sp.
0
8
16
MAGs per cluster
161 MAGs in 38 clusters;
shown: 135 MAGs (84%)
Cofactors & vitamins
Lipid
Amino acid
Carbohydrate
Energy
Nucleotide
M00015 M00016 M00017 M00018 M00019 M00021 M00022 M00028 M00526 M00527 M00570 M00844 M00001 M00002 M00003 M00004 M00005 M00007 M00307 M00549 M00909 M00157 M00579 M00082 M00083 M00093 M00120 M00140 M00048 M00049 M00050 M00051 M00052 M00053 M00938
Module completeness (%)
0
50
100